

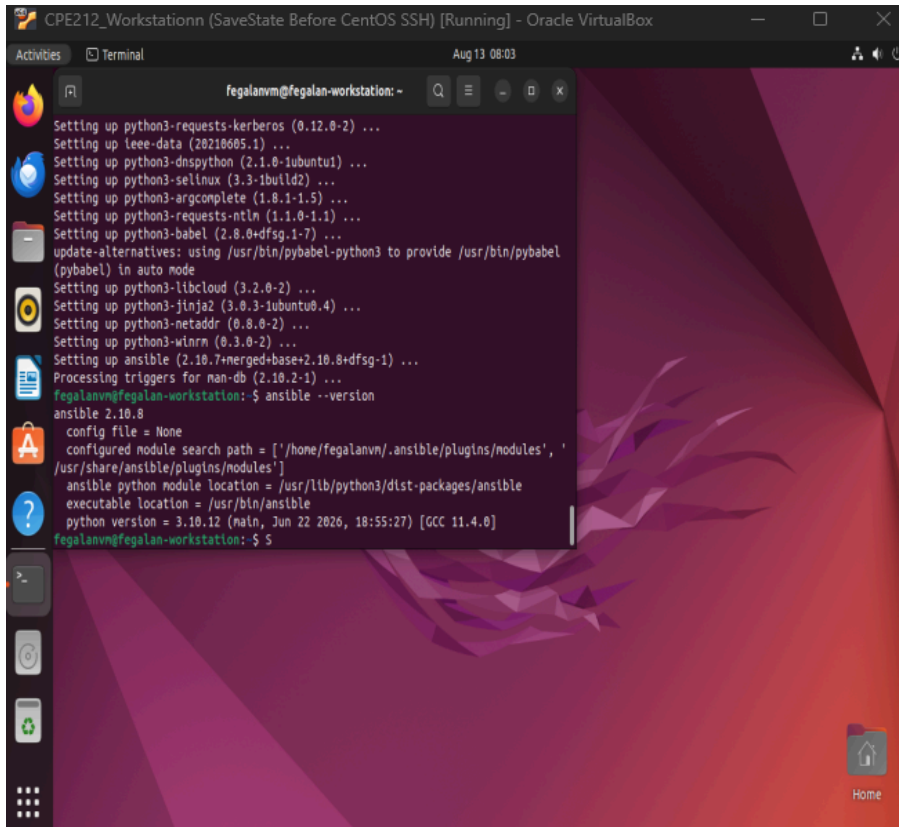
COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	08/13/2026	DATE SUBMITTED:	08/13/2026
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PRELIM SKILLS EXAM

1. PROCEDURE OUTPUT

Hostname	Operating System	IP Address
ubuntu1	Ubuntu	192.168.56.118
ubuntu2	Ubuntu	192.168.56.119
centos1	CentOS	192.168.56.117

PART A:



```

CPE212_Workstationn (SaveState Before CentOS SSH) [Running] - Oracle VirtualBox
Aug 13 08:03
fegalanvm@fegalan-workstation: ~
Setting up python3-requests-kerberos (0.12.0-2) ...
Setting up leee-data (20210605.1) ...
Setting up python3-dnspython (2.1.0-1ubuntu1) ...
Setting up python3-selinux (3.3-1build2) ...
Setting up python3-argcomplete (1.8.1-1.5) ...
Setting up python3-requests-ntlm (1.1.0-1.1) ...
Setting up python3-babel (2.8.0+dfsg.1-7) ...
update-alternatives: using /usr/bin/pybabel-python3 to provide /usr/bin/pybabel
(pybabel) in auto mode
Setting up python3-libcloud (3.2.0-2) ...
Setting up python3-jinja2 (3.0.3-1ubuntu0.4) ...
Setting up python3-netaddr (0.8.0-2) ...
Setting up python3-winrm (0.3.0-2) ...
Setting up ansible (2.10.7+merged-base+2.10.8+dfsg-1) ...
Processing triggers for man-db (2.10.2-1) ...
fegalanvm@fegalan-workstation: $ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/fegalanvm/.ansible/plugins/modules',
  /usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
fegalanvm@fegalan-workstation: $
  
```

```
fegalanvm@fegalan-workstation:~$ ssh fegalanvm@192.168.56.118
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

71 updates can be applied immediately.
71 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 09:08:46 2026 from 192.168.56.111
fegalanvm@fegalan-server1:~$
```

```
fegalanvm@fegalan-workstation:~$ ssh fegalanvm@192.168.56.119
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

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To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 09:08:46 2026 from 192.168.56.111
fegalanvm@fegalan-server2:~$
```

```
fegalanvm@fegalan-workstation:~$ ssh centos_chris@192.168.56.117
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 08:15:12 2026
[centos_chris@localhost ~]$
```

PART B:

```
fegalanvm@fegalan-workstation: ~/ansible-lab
GNU nano 6.2 inventory.yaml
all:
  children:
    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.118
        ubuntu2:
          ansible_host: 192.168.56.119
    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.117

Read 14 lines
Help Write Out Where Is Cut Execute Location Undo Set Mark
Exit Read File Replace Paste Justify Go To Line Redo Copy
```

```

fegalanvm@fegalan-workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
  "_meta": {
    "hostvars": {
      "server1": {
        "ansible_host": "192.168.56.118"
      },
      "server2": {
        "ansible_host": "192.168.56.119"
      },
      "server3": {
        "ansible_host": "192.168.56.117",
        "ansible_user": "centos_chris"
      }
    }
  },
  "all": {
    "children": [
      "centos",
      "linuxservers",
      "ubuntu",
      "ungrouped"
    ]
  },
  "centos": {
    "hosts": [
      "server3"
    ]
  },
  "linuxservers": {
    "children": [
      "centos",
      "ubuntu"
    ]
  },
  "ubuntu": {
    "hosts": [
      "server1",
      "server2"
    ]
  }
}

```

```

fegalanvm@fegalan-workstation:~/ansible-lab$ ansible all --list-hosts
hosts (3):
  192.168.56.118
  192.168.56.119
  192.168.56.117

```

```

fegalanvm@fegalan-workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature will
be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}

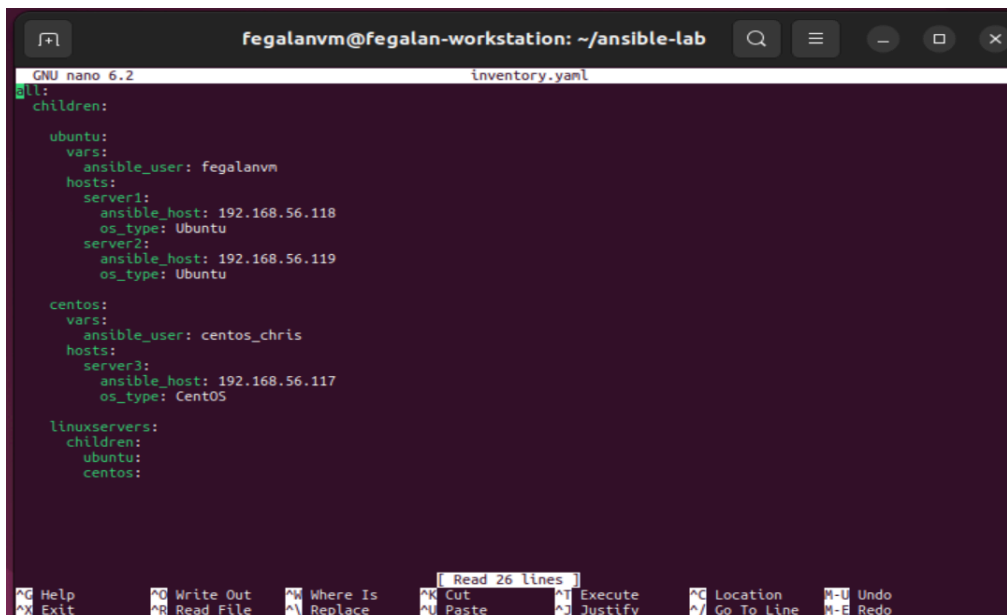
```

PART C:

```
fegalanvm@fegalan-workstation:~/ansible-lab$ ansible linuxservers -n ping -i inventory.yaml
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference\_appendices/interpreter\_discovery.html for more information. This feature will
be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
fegalanvm@fegalan-workstation:~/ansible-lab$

fegalanvm@fegalan-workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.yaml
hosts (3):
  server1
  server2
  server3
fegalanvm@fegalan-workstation:~/ansible-lab$
```

PART D:



```
fegalanvm@fegalan-workstation: ~/ansible-lab
GNU nano 6.2 inventory.yaml
children:
  ubuntu:
    vars:
      ansible_user: fegalanvm
    hosts:
      server1:
        ansible_host: 192.168.56.118
        os_type: Ubuntu
      server2:
        ansible_host: 192.168.56.119
        os_type: Ubuntu
  centos:
    vars:
      ansible_user: centos_chris
    hosts:
      server3:
        ansible_host: 192.168.56.117
        os_type: CentOS
  linuxservers:
    children:
      ubuntu:
      centos:
```

```
fegalanvm@fegalan-workstation: ~/an
GNU nano 6.2 testvars.yaml
- name: Display Variables
  hosts: linuxservers
  tasks:
    - name: Show Operating System
      debug:
        msg: "Host {{ inventory_hostname }} is running {{ os_type }}"

fegalanvm@fegalan-workstation:~/ansible-lab$ nano testvars.yaml
fegalanvm@fegalan-workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to
using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False
in ansible.cfg.
ok: [server3]

TASK [Show Operating System] *****
ok: [server2] => {
  "msg": "Host server2 is running Ubuntu"
}
ok: [server1] => {
  "msg": "Host server1 is running Ubuntu"
}
ok: [server3] => {
  "msg": "Host server3 is running CentOS"
}

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

PART E:

```
fegalanvm@fegalan-workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to
using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False
in ansible.cfg.
ok: [server3]

TASK [Show Operating System] *****
ok: [server2] => {
  "msg": "Host server2 is running Ubuntu"
}
ok: [server1] => {
  "msg": "Host server1 is running Ubuntu"
}
ok: [server3] => {
  "msg": "Host server3 is running CentOS"
}

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
fegalanvm@fegalan-workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to
using the discovered platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False
in ansible.cfg.
ok: [server3]

TASK [Ping all managed ndoes] *****
ok: [server2]
ok: [server3]
ok: [server1]

TASK [Display Hello World] *****
ok: [server1] => {
  "msg": "Hello World from server1"
}
ok: [server2] => {
  "msg": "Hello World from server2"
}
ok: [server3] => {
  "msg": "Hello World from server3"
}

TASK [Display Operating System] *****
ok: [server1] => {
  "msg": "Operating System: Ubuntu"
}
ok: [server2] => {
  "msg": "Operating System: Ubuntu"
}
ok: [server3] => {
  "msg": "Operating System: CentOS"
}

PLAY RECAP *****
server1      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

```
fegalanvm@fegalan-workstation: ~/ansible-lab  GNU nano 6.2  challenge.yaml
name: Supplementary Activity
hosts: servers

tasks:
  - name: Display Message
    debug:
      msg: "Welcome to Ansible Laboratory"

[ Read 7 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify   ^_ Go To Line
```

```

fegalanvm@fegalan-workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Supplementary Activity] *****

TASK [Gathering Facts] *****
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using /usr/bin/python for
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/referen
This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.c
ok: [server3]
ok: [server1]

TASK [Display Message] *****
ok: [server1] => {
  "msg": "Welcome to Ansible Laboratory"
}
ok: [server3] => {
  "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
  
```

Guide Questions

1. What is the role of the Control Node in Ansible?

- In ansible the control node is where you execute commands on remote servers. It basically acts as the main hub for the servers connected to it. Where executing playbook commands will run on groups of servers depending on the inventory node.

2. Why is an inventory file required?

- The inventory file sets groups, metagroups, and variables that allow for flexibility in terms of running commands on your different servers. Let's say you want to ping server 1 and server 3 without pinging server 2 you could do that but also it allows for you to specify names of variables like the one with msg and "{%}" where you can do variable substitutions which makes it easier and more convenient to tell and connect to these servers.

3. What is the purpose of a metagroup?

- It is a convenient way of combining multiple groups into a single variable. It allows group servers like lets say you want to combine hr and accounting to one group but you don't want to include resources to the group for specific reasons, metagroups allow you to simplify calling many groups into one without affecting other groups. It also simplifies the playbook by a lot making it more understandable and readable to the developer who would read it.

4. How do variables simplify Ansible administration?

- Variables allow for functions like debug msg to do variable substitutions so that you don't have to call each server to do one function for a million lines which takes time and is very slow which is something we don't want in real-time servers that handles a bunch of traffic.

5. What is the difference between an ad hoc command and a playbook?

- The difference is that ad hoc commands are temporary one line commands that don't get saved permanently and are not automated meanwhile playbook is a file that contains a set of commands that can be reused, edited, and saved for automation.

CONCLUSION

- Using ansible was very interesting handling virtual servers and creating playbook commands. Going through the setting up of ssh to ansible and creating the directory of /etc/ansible and file of hosts to declare the groups each server ip belongs in was not specified within the activity but something I realized as I was going through this activity because of the specific commands that was tasked to run within it. Overall, I've learned about automation with ansible and I could see some use cases for it for my everyday tasks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.